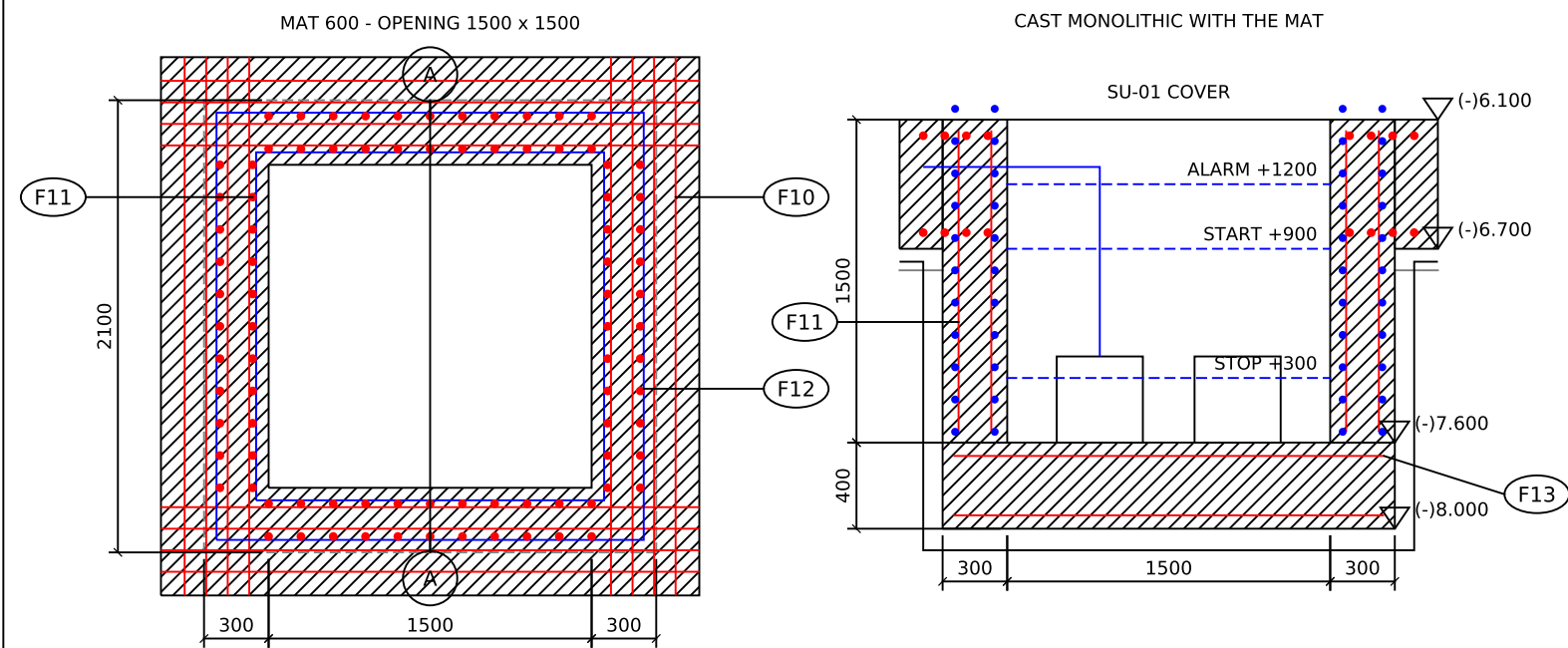
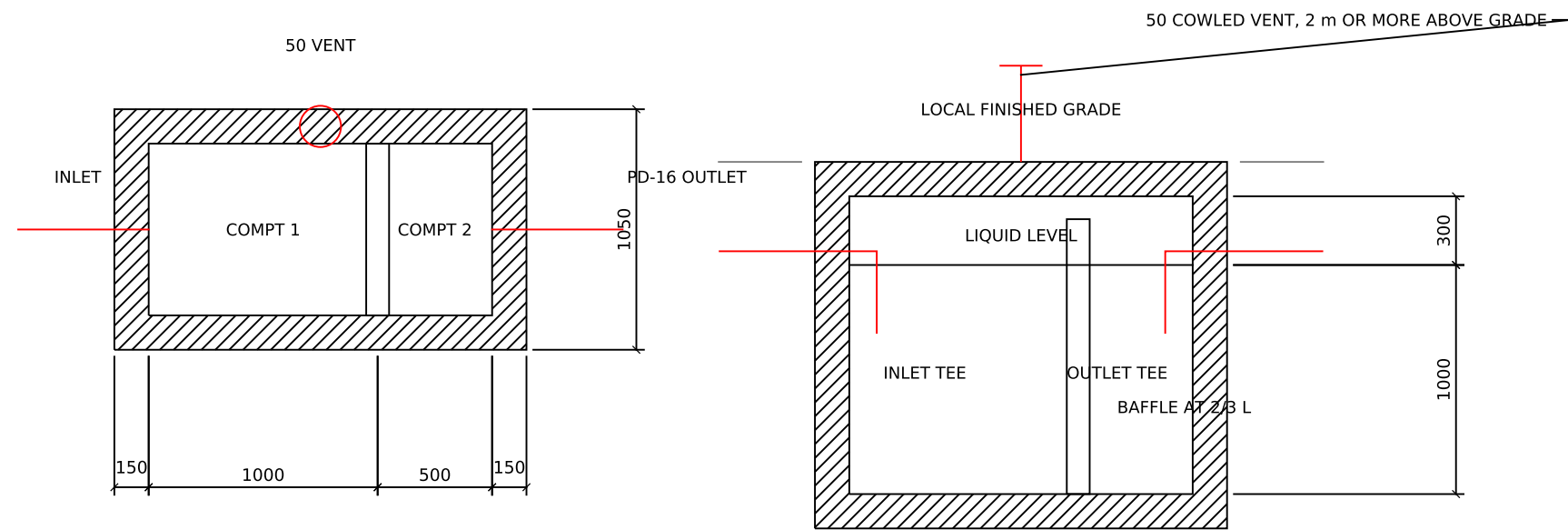




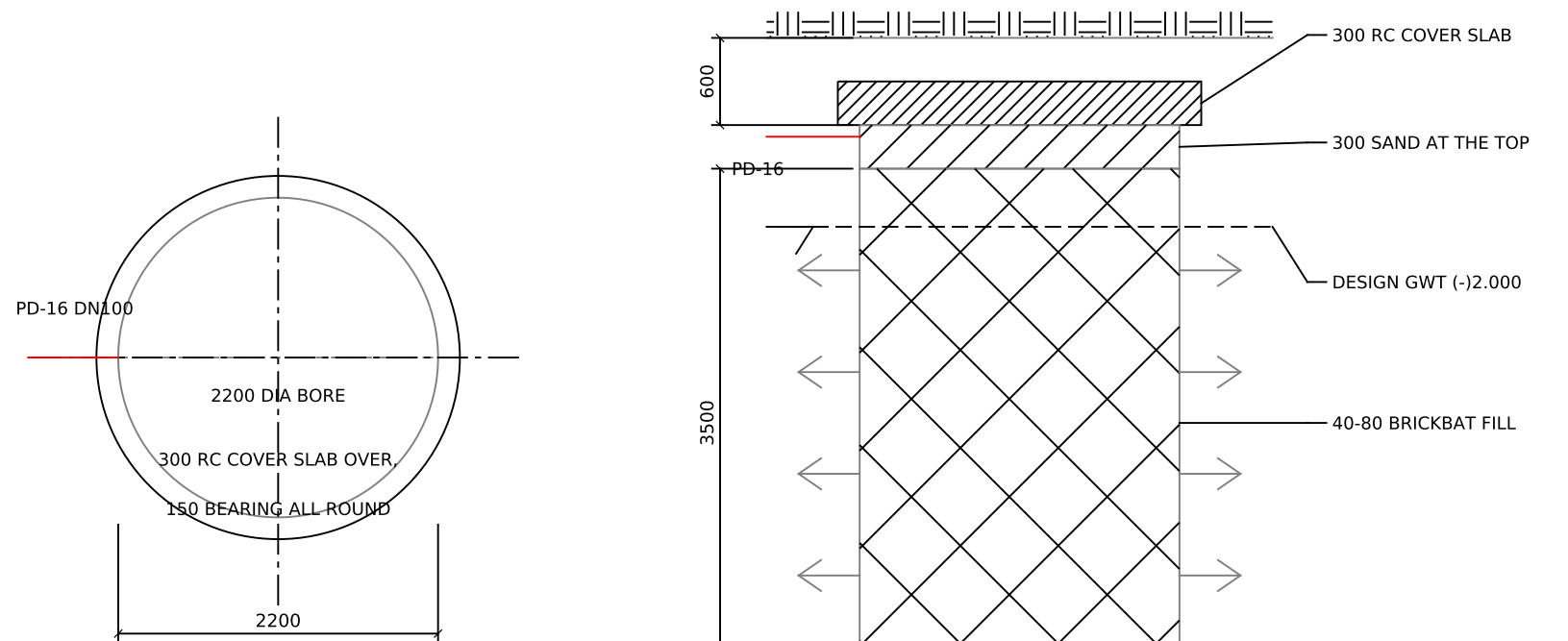
1 SUMP PIT SU-01 - REINFORCEMENT PLAN 1 : 35 BAY 5, X 11068 - 12568, Y 900 - 2400

2 SUMP PIT SU-01 - SECTIONAL ELEVATION A-A 1 : 35 3.375 m3, INVERT (-)7.600



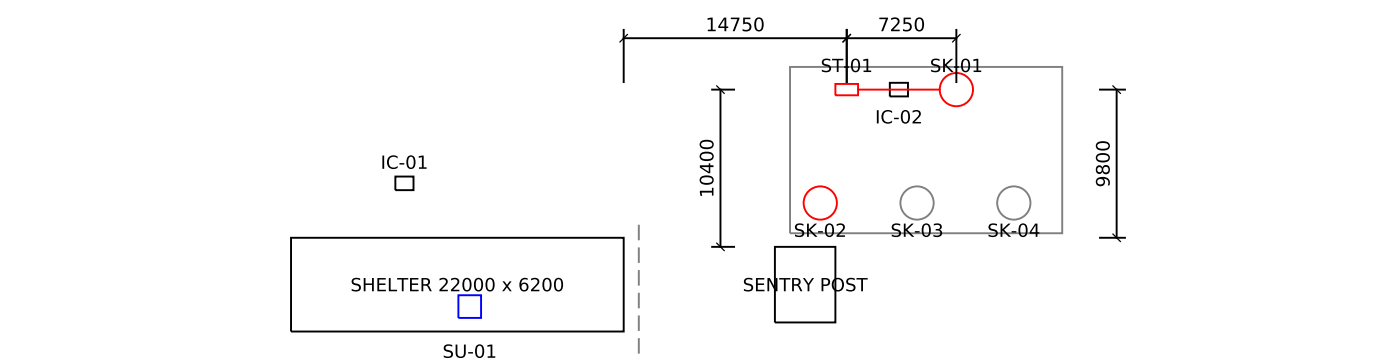
3 SEPTIC TANK ST-01 - PLAN 1 : 30 1500 x 750 LIQUID, 150 WALLS

4 SEPTIC TANK ST-01 - SECTIONAL ELEVATION 1 : 30 IS 2470 (Pt 1):1985



5 SOAK PIT SK-01 - PLAN 1 : 50 SK-02 IS IDENTICAL

6 SOAK PIT SK-01 - SECTIONAL ELEVATION 1 : 50 2200 DIA x 3500 EFFECTIVE, IS 2470 (Pt 2):1985



7 KEY PLAN - EXTERNAL WORKS LOCATION 1 : 500 EAST OF THE SHELTER AND NORTH OF THE SENTRY POST

SUMP, PUMP AND TANK SCHEDULE				
TAG	ITEM	DUTY / SIZE	LEVEL	NOTES
SU-01	CLEAN SUMP	1500 x 1500 x 1500 = 3.375 m3	(-)7.600	BAY 5. WALLS 300, BASE 400, BASE SLAB (-)8.000
PU-01	SUBMERSIBLE PUMP	1.5 L/s	(-)7.600	DUTY, AUTO-ALTERNATING WITH PU-02. AT (11468, 1650)
PU-02	SUBMERSIBLE PUMP	1.5 L/s	(-)7.600	STANDBY. AT (12168, 1650)
PU-03	HAND PUMP	MANUAL	(-)6.100	THIRD LINE OF DEFENCE, INDEPENDENT OF POWER
PUMP CONTROL, ABOVE THE INVERT: START +900 STOP +300 HIGH-LEVEL ALARM +1200				
SU-02	STAIRWELL SUMP	1.0 m3	(-)2.000	AT THE PLATFORM
PU-04	STAIRWELL PUMP	2 L/s	(-)2.000	DUTY
PU-05	STAIRWELL PUMP	2 L/s	(-)2.000	STANDBY. NON-RETURN VALVE ON THE RISING MAIN
TK-01	DECON EFFLUENT TANK	1000 L	(-)6.100	BAY 8. TANKER ONLY - NEVER TO THE CLEAN SUMP

STRUCTURAL ELEMENT SCHEDULE						
ELEMENT	THK	SIZE	COVER	d	REINFORCEMENT	NOTES
SU-01 SUMP PIT WALLS	300	1500 x 1500 CLEAR	50	244	T16 @ 150 EF EW	CAST MONOLITHIC WITH THE MAT
SU-01 SUMP PIT BASE	400	2100 x 2100	50	244	T16 @ 150 EF EW	BASE SLAB (-)8.000
MAT OPENING OVER SU-01	600	1500 x 1500	75 / 50	517	4-T20 TRIMMERS EF / SIDE	Ld 800 BEYOND THE OPENING EACH WAY
ST-01 SEPTIC TANK	150	1500 x 750 LIQUID	-	-	TO THE STR ENGINEER'S DETAIL	TWO COMPARTMENTS, BAFFLE AT 2/3 L
SK-01 / SK-02 COVER SLAB	300	2500 DIA	-	-	TO THE STR ENGINEER'S DETAIL	U/S 600 BELOW LOCAL FINISHED GRADE
IC-01 / IC-02 CHAMBER	150	600 x 450 CLEAR	-	-	TO THE STR ENGINEER'S DETAIL	BENCH TO THE CHANNEL INVERT

BAR MARK SCHEDULE - SUMP PIT SU-01					
MARK	BAR	SPACING	LOCATION	CUT mm	No.
F10	T20	-	SUMP-PIT OPENING TRIMMERS	3100	32
F11	T16	150	VERTICAL, EACH FACE	2300	96
F12	T16	150	HORIZONTAL RINGS, EACH FACE	8000	22
F13	T16	150	BOTH WAYS, EACH FACE	2000	56

EXTERNAL DRAINAGE STRUCTURE SCHEDULE				
TAG	ITEM	SIZE	CENTRE (X, Y)	NOTES
ST-01	SEPTIC TANK	1500 x 750 x 1000 LIQUID = 1125 L	(36750, 16000)	10 USERS AT 45 L/DAY. FREEBOARD 300, OVERALL 1300
SK-01	FOUL SOAK PIT	2200 DIA x 3500 EFFECTIVE	(44000, 16000)	24.19 m2 SIDE AREA. TAKES THE ST-01 EFFLUENT
SK-02	STORM SOAKAWAY	2200 DIA x 3500 EFFECTIVE	(35000, 8500)	TAKES THE CLEAN SUMP DISCHARGE AND THE ENTRY CHANNEL
SK-03	STAIRWELL SOAKAWAY	-	(41400, 8500)	FOOTPRINT RESERVED. TAKES THE SU-02 RISING MAIN
SK-04	HEADHOUSE SOAKAWAY	-	(47800, 8500)	FOOTPRINT RESERVED
IC-01	INSPECTION CHAMBER	600 x 450	(7500, 9800)	ON PD-11 AT THE CHANGE OF DIRECTION
IC-02	INSPECTION CHAMBER	600 x 450	(40200, 16000)	ON PD-16, FOR SEPTIC TANK DE-SLUDGING ACCESS

EXTERNAL PIPE SCHEDULE					
REF	SERVICE	FROM -> TO	DN	GRADIENT	LENGTH
PD-16	FOUL	ST-01 -> SK-01	DN100	1:100	5.40 m
PD-06	RISING MAIN	SERVICE ENTRY PLATE -> SK-02	DN50	PUMPED	27.00 m
PD-11	STORM	CP-10 CATCHPIT -> SK-02	DN100	1:100	32.35 m
PD-13	RISING MAIN	SU-02 STAIRWELL SUMP -> SK-03	DN50	PUMPED	29.60 m

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA

NOTES

- ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS. WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
- THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH, STR AND SERVICES DRGS.
- SUMP PIT SU-01: CONCRETE M35, w/c NOT MORE THAN 0.45. CAST MONOLITHIC WITH THE MAT. REINFORCEMENT Fe500D TO IS 1786:2008, Ld 40 PHI, LAPS 50 PHI STAGGERED SO THAT NOT MORE THAN 50 % ARE SPLICED AT ONE SECTION.
- THE TANKING MEMBRANE IS DRESSED AROUND SU-01 AND IS CONTINUOUS WITH THE MAT TANKING. NO BAR MAY BE DISPLACED TO CLEAR A SERVICE.
- THE SU-01 DISCHARGE CARRIES AN ISOLATION VALVE, A GAS-TIGHT NON-RETURN VALVE, A BLAST CHECK VALVE AND A 75 DEEP-SEAL TRAP IN SERIES, ALL INSIDE THE GAS-TIGHT ENVELOPE.
- SEPTIC TANK ST-01 TO IS 2470 (Pt 1):1985 AND SOAK PIT SK-01 TO IS 2470 (Pt 2):1985. THE SOAK PIT SIDE AREA ALONE IS COUNTED - THE BASE CLOGS.
- ST-01 AND SK-01 WALL, BASE AND COVER-SLAB REINFORCEMENT IS TO THE STRUCTURAL ENGINEER'S DETAIL AND IS NOT SCHEDULED ON THIS SHEET.
- A PERCOLATION TEST TO IS 2470 (Pt 2) Cl. 4 IS TO BE CARRIED OUT AT THE POSITION OF EACH PIT, AT BOTH THE PIT INVERT AND THE TRENCH INVERT, BEFORE THE PITS ARE BUILT.
- CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

SEPTIC TANK, SOAK PIT & SUMP PIT - PLANS, SECTIONS & SCHEDULES

CBRN HARDENED UG OPS ROOM

SHEET 11

Project number	01
Date	16 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI
MEP011	
Scale	As indicated